

Project III

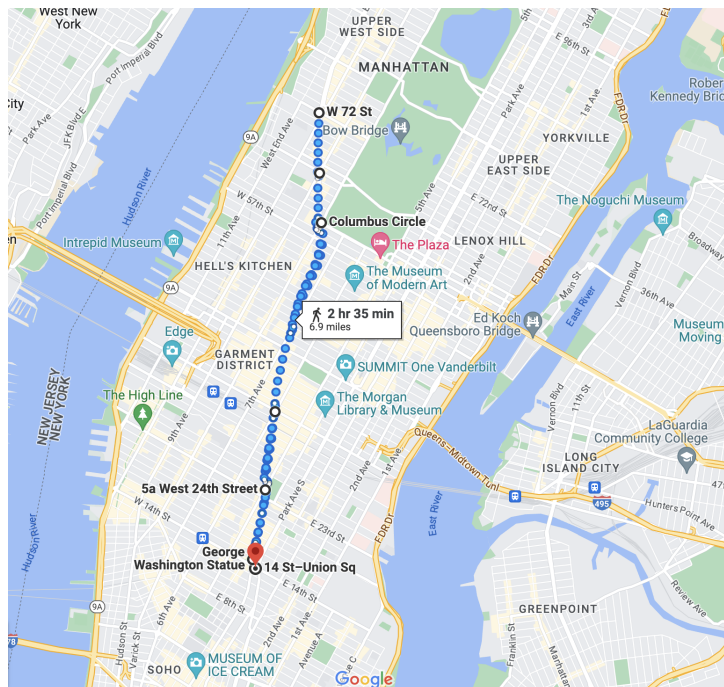
Let's do some New York mathematics!

Part I

Part of New York City is set up on a grid where streets run horizontally and avenues run vertically. But Broadway is the only road in Manhattan that runs diagonally cutting the streets and avenue from 14th street to 72nd street. We will use this fact to develop a linear regression line for Broadway in New York City.

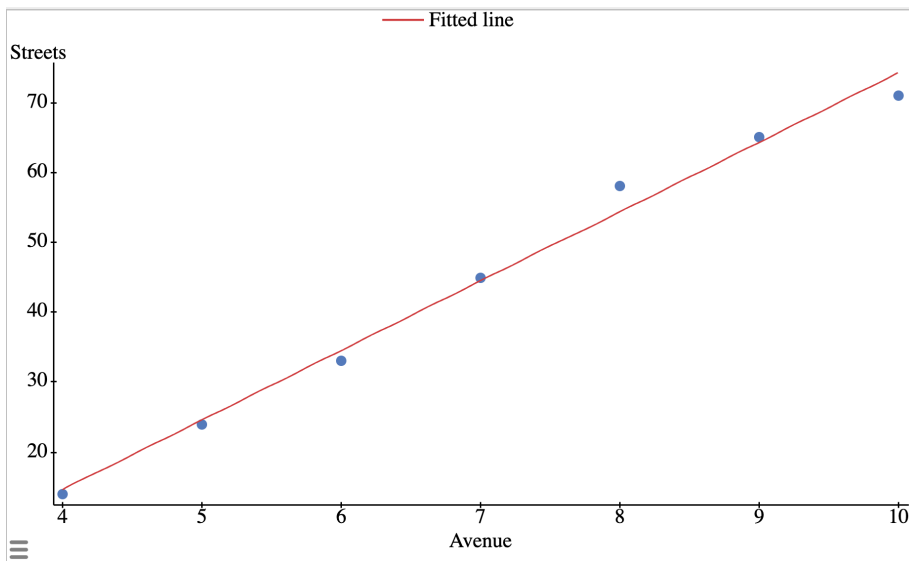
First of all, we need a grid. Since there is no 0th avenue and 0th street, the corner of Avenue A and East 1st street is the origin of the grid. Also, the avenues decrease as we move from left to right which is unconventional, but it does the job.

Broadway runs parallel to avenues up to 4th avenue, so we will start our data collection from 4th avenue and 14th street. On our x -axis we will plot the avenues and on y -axis we will plot the streets. Broadway meets avenue and street on 4th avenue and 14th street, so first data point is $(4, 14)$. Similarly, Broadway meets the intersection of 5th avenue and 24th street. Hence, our second data point is $(5, 24)$. Broadway meets the intersection of 6th avenue and 33rd street. Thus, our third data point is $(6, 33)$. Using similar methodology, we have the rest of our observations as $(7, 45)$, $(8, 58)$, $(9, 65)$, and $(10, 71)$.



You can verify this result yourself via Google Maps or Apple Maps before proceeding to the next part.

Now we will plot this information on a scatterplot and create a line of best fit for Broadway. The linear equation is calculated as $\text{Streets} = -25.2 + 9.93 \times \text{Avenue}$. The r^2 value is 0.99



Now let's answer the following questions:

- Interpret the slope of the regression line.
- Interpret the y -intercept of the the regression line. Does it make sense in real life?
- Suppose you do not have street number, but you partial information that that you need to reach Broadway in between 8th and 9th avenue (8.5th avenue for mathematical sense). Using the regression line, predict in which street you would expect to see the Broadway in between 8th and 9th avenue. Verify your prediction using maps. How close was the prediction?
- Was the prediction in question 3 a good prediction or bad prediction. Explain.
- How well the does the data fit the regression model in our least square regression model? Write in terms of percentage.

Part II

This part of the project simulates a dataset representing various factors influencing Broadway productions. You'll analyze the relationships between different variables to determine meaningful insights through linear regression.

The Broadway dataset (find on Canvas) contains **35 rows and 11 columns** representing metrics such as ad spend, ticket sales, rent costs, and social media engagement for Broadway shows. Your task is to explore which variables impact profitability or other outcomes and identify both suitable and unsuitable models.

1. Create scatterplots with lines of best fit and generate the corresponding linear equations between:
 - **Ad_Spend vs. Profit**
 - **Ticket_Sales vs. Profit**
 - **Online_Views vs. Profit**
 - **Non_Linear_Feature vs. Profit**
- Which independent variable seems to have the strongest correlation with **Profit**?
- Are there any scatterplots that indicate non-linear behavior?
- Why is it essential to identify non-linear relationships before performing linear regression?
2. Choose two variables from the dataset you believe will have a meaningful linear relationship with **Profit** or **Ticket_Sales**.
3. Run linear regression on your selected variables.
4. Create residual plots for each linear regression model.
 - What patterns (if any) do you observe in the residual plots?
 - How does the residual plot help confirm or reject the suitability of your model?
 - Why might some variables, such as **Day_of_Week**, be unsuitable for linear regression?
5. Calculate the R^2 values for your regression models.
6. Interpret the R^2 values:
 - What proportion of the variance in **Profit** or **Ticket_Sales** is explained by the independent variable(s)?
 - If the R^2 value is low, what does this tell you about the model?
 - Are the variables you analyzed practical predictors of profitability? Why or why not?